

CS228 Logic for Computer Science

Exercise Problem Set 1

Part 1

1. Draw parse trees for the following propositional formulae.
 - (a) $(p \wedge \neg q) \rightarrow (p \vee q)$,
 - (b) $(p \rightarrow q) \wedge (\neg p \wedge \neg q)$,
 - (c) $(p \wedge \neg((\neg q) \rightarrow (r \vee \neg s)))$,
 - (d) $(\neg(p \vee q)) \wedge (r \wedge (r \rightarrow p))$.
2. For each of the parts in the above question, give a satisfying assignment (i.e. assign truth values to the propositions such that the formula evaluates to true), if it exists. For parts (a) and (b) above, draw their truth-tables, and use them to answer if the respective formula are valid (i.e., every assignment evaluates to true).
3. Encode the following two statements into propositional logic formulae:
 - (a) If it rains today, the road will get wet.
 - (b) The road today will get wet, or it did not rain.
 - (c) If the road does not get wet, it did not rain today.

Check the truth tables for these 3 formulae. What do you notice?

4. There was a course assignment and there are three suspects for plagiarism: Shivansh, Abhay, and Shishir. You overheard the following statements:
 - Shivansh says: “If Abhay did it, then Shishir helped him.”
 - Abhay says: “I didn’t do it. Also I am not lying.”
 - Shishir says: “I’m innocent, and Shivansh is telling the truth.”

Only one person has plaggd. Assume only the guilty person is lying and the two innocent ones are telling the truth. The issue escalated to DADAC and now you are tasked with the duty to determine who committed the plagiarism. Encode the statements into propositional logic and deduce the answer.

Note. If you help someone in plag, you are also guilty! Plagiarism has terrible consequences! What can happen to someone who plags: <https://www.iitb.ac.in/newacadhome/punishments201521July.pdf>

5. Given three propositions ϕ , ψ , and ξ , prove the following properties by writing their Truth Tables:
 - (a) $\phi \rightarrow \psi \equiv \neg\phi \vee \psi$,
 - (b) $\phi \wedge \psi \equiv \neg(\neg\phi \vee \neg\psi)$,
 - (c) $\phi \vee (\psi \wedge \xi) \equiv (\phi \vee \psi) \wedge (\phi \vee \xi)$,
 - (d) $\phi \oplus (\psi \oplus \xi) \equiv (\phi \oplus \psi) \oplus \xi$.

Part 2

6. Let F , G , and H be propositional formulas, and let S be a set of formulas. Determine which of the following statements are true. Justify your answers.
- (a) If F is unsatisfiable, then $\neg F$ is valid.
 - (b) If $F \rightarrow G$ is valid and F is satisfiable, then G is satisfiable.
 - (c) $P_1 \rightarrow (P_2 \rightarrow (P_3 \rightarrow \dots (P_n \rightarrow P_1) \dots))$ is valid.
 - (d) $S \models F$ and $S \models \neg F$ cannot both hold.
 - (e) If $S \models F \vee G$, $S \cup \{F\} \models H$ and $S \cup \{G\} \models H$, then $S \models H$.

7. There are n switches and m lamps. For each $i \in \{1, \dots, n\}$, let the atomic proposition S_i denote that switch i is ON. For each $j \in \{1, \dots, m\}$, let the atomic proposition L_j denote that lamp j is ON.

The circuit behaves according to the following rules:

- Lamp 1 is ON if and only if exactly one of the switches is ON.
- Lamp 2 is ON if and only if an even number of switches are ON.
- For each $j \in \{3, \dots, m\}$, lamp j is ON if and only if switch $j - 2$ is ON (assume $m \leq n + 2$).

Write propositional logic formulas that express each of the following:

- (a) The behaviour of Lamp 1. (evaluates to True if Lamp 1 is ON).
 - (b) The behaviour of Lamp 2. (evaluates to True if Lamp 2 is ON).
 - (c) All lamps are ON.
 - (d) Exactly one lamp is ON.
8. A set of connectives is called *universal* if for every Boolean function there is an equivalent formula with only connectives from that set. For example, we saw in class that $\{\neg, \vee\}$ is universal for propositional logic since any occurrence of $\wedge$ and $\rightarrow$ can be removed using the equivalences:

$$\begin{aligned}\phi \rightarrow \psi &\equiv \neg\phi \vee \psi \\ \phi \wedge \psi &\equiv \neg(\neg\phi \vee \neg\psi)\end{aligned}$$

- (a) The binary connective nand, written $F \downarrow G$, is defined by the truth table corresponding to

$$F \downarrow G \equiv \neg(F \wedge G).$$

Prove that nand is universal.

- (b) Consider the ternary logical connective ITE (If-Then-Else), defined semantically as

$$\text{ITE}(p, q, r) \equiv (p \wedge q) \vee (\neg p \wedge r).$$

Prove or disprove: the set of connectives $\{\text{ITE}, \top, \perp\}$ is universal.

- (c) Prove or disprove: the set $\{\wedge, \vee, \rightarrow\}$ is universal.

9. The *Pigeon Hole Principle* states that if there are $n + 1$ pigeons sitting amongst n holes then there is at least one hole with more than one pigeon sitting in it. For $i \in \{1, 2, \dots, n + 1\}$ and $j \in \{1, 2, \dots, n\}$, let the atomic proposition $P(i, j)$ indicate that the i -th pigeon is sitting in the j -th hole.

Write out a propositional logic formula that states the Pigeon Hole Principle.

10. A *perfect matching* in an undirected graph $G = (V, E)$ is a subset of edges $M \subseteq E$ such that every vertex $v \in V$ is an endpoint of exactly one edge in M . Given a finite graph G , describe how to obtain a propositional logic formula φ_G such that φ_G is satisfiable iff G has a perfect matching. The length of your formula should not grow exponentially, so avoid simply listing all possible edge combinations.